

Week 1 Summary Report

Participant Name: Mukesh Kumar C

Roll No:251090050560

Domain: Software and ML

Repository Link:

[https://github.com/MukeshKumarC-0309/Syner
gy_TP.git](https://github.com/MukeshKumarC-0309/Synergy_TP.git)

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Abstract

This report gives a short summary of the all the work done in tasks 1-5. It iterates through every task covering the key lessons learned, the workflow which I followed to achieve the desired result, the libraries used, the code, and the lessons I learnt throughout this process. This report will also address key details regarding the need for certain practices which makes for clean code creation, clean output generation and properly maintaining code in global servers accessible to everyone making it a collaborative process. It traverses step by step starting from setting up virtual environments, basic python scripting , Json dumping, csv parsing , data cleaning , data visualization , and proper version control and submission of code.

Introduction

Task-1

Task 1 focuses on setting up our GitHub repos, integrating them with IDE for more accessible coding, setting up virtual environments, basic python scripting and running them through the terminal.

Task-2

Task 2 focuses on file handling concepts and how to write data into a Json file using the python's in-built libraries.

Task-3

Task 3 focuses on data parsing as we build both a manual and pandas-based scripting model to check the efficiency and ease of coding for both methods.

Task-4

Task 4 focuses on cleaning messy data using the pandas module to achieve a uniform spread of data by handling the outliers/incomplete data.

Task-5

Task 5 focuses on utilizing the cleaned code to the next level by incorporating visualization techniques by using in built methods of matplotlib.

Development Environment and Version Control

GitHub is a platform which helps incorporate git to your projects which also helps you store your projects elsewhere. It is a tool for version control, which involves tracking the changes in our code. Every time we make a change in our code, we must update it using git so as to maintain a proper record of the work we have done over a period of time.

Repository Hygiene refers to the proper practices followed so as to maintain a clean record of the changes we have made to our code or the new projects we have added over time. A few practices include using clean “commit messages” so that in the future, other users(including self) can understand the process that went into the project. We must also maintain proper folder structure such that accessing a small part of the entire project would be easy as everything would be marked properly and would follow a proper sequence. We must also try to avoid unwanted files such as pycache/venv files as it would be cluttered causing confusion. We can achieve this using the .gitignore file which we will talk about in the later parts. Finally another practice which would help repository hygiene is creation of README files at the beginning of every new project. This would provide ample information to the users accessing the project about the necessary tools and workflow of the project which would allow future users to easily access our code.

Virtual Environment is mainly used when different projects require different versions of libraries. The problem with installing libraries normally is that they are deployed globally which means that handling different versions would be confusing and even difficult at times. Virtual Environments acts as a small box isolated from the device where we could maintain a different version in different environments. This also helps in having a clear difference for different

projects instead of having a common space to install libraries. It is a good practice to maintain 1 virtual environment per project.

Requirements.txt is an extension of the concept used in virtual environments. It is used to store the different libraries along with their version installed in a specific virtual environment. Anyone who wishes to use our project for future use, while cloning the repo can just access the required data through “pip install -r requirements.txt” which would have all the libraries used in it.

.gitignore is a file which specifies the files and folders which are to be ignored by GitHub. This is particularly useful to avoid storing temporary files such as venv and pycache which would disrupt the clean structure of our code. It must also avoid .env files as it would contain confidential information which would compromise the identity of the user.

Python and File Handling Concepts

Various Python functions are used throughout the tasks ranging from the basic print() statement to sys.argv() statements which help set up argument vectors while running the code.

Functions-This broke down the entire code into smaller chunks of block code which makes readability and reusability much simpler as this allows us to call the function any number of times.

List and Dictionaries-These are data lists which help store data together allowing us to read them better.

Loops-Loops are used to iterate through the process saving us precious time instead of writing the same code block ‘n’ number of times to achieve the desired result.

String Methods-Various String Methods are used throughout the tasks such as .strip(), .replace(), .split() etc. which helps in better string handling allowing us to manipulate the given data to achieve required results

os.makedirs()-This is a crucial function which checks if the specified directory exists. If not, it creates a directory immediately in its place which removes the fact that a missing inventory would've caused an error causing the program to terminate.

CSV Parsing and Pandas

Manual Parsing is used to read a csv file and display its contents without using any in-built libraries but rather use basic python functions to separate the values and display them.it involves invoking the said csv file, separating the headers first using .split() function and then iterating through every row separating the values by ',' and storing them in a dictionary thereby creating a list of dictionary. This allows for us to view the information better and manipulation of said data would be much easier as dealing with lists and dictionaries is much simpler due to the wide assortments of functions available for said data manipulation.

Then we switched over to a more niche and function-based parser using the pandas libraries which provides more functions which allow for data parsing of csv files with just one function(pd.read_csv) which automatically separates the values and stores in a data frame(a tabular structure) which allows for better viewing and manipulation of said data.

Type Conversion is an important concept which we learnt throughout these tasks as there is no guarantee that the data we receive will have proper and structured data. Hence it is important to go through the data and make necessary changes for a more uniform structure of data. Here, type conversion plays an important role as most of the times, we would require converting the numerical categories to int which would allow for us to manipulate and calculate other factors using data. Hence type conversion is essential to perform the required operations on the dataset provided to us to generate a more precise result.

Summary generation refers to using the cleaned data and producing a structured report with all key features/statistics such as count, average etc. In a manual parser we had to calculate each

statistic manually using loops and basic arithmetic operations, however in the pandas parser it was much easier to group data based on a header and calculate their sum/avg etc. The purpose of generating a summary before and after cleaning in Task 4 is to show what changed. The before summary captures how many duplicates, missing values, and inconsistent domains existed. The after summary confirms they were all resolved. The entire workflow involved importing raw csv file, parsing through python, type conversion(if necessary) and summary generation.

Data Cleaning

Data cleaning refers to practice of maintaining proper code and dealing with outliers such as missing data, inconsistent values, type fixes, handling duplicates and data normalization. We achieve this practice by utilizing the pandas functions.

Missing Values are handled by either dropping them or in our case, replacing them with the mean/median/mode of that said column. For example, in our code we used `.fillna()` function along with `df['score'].mean()` because it's better to handle that column using the mean score achieved by the students to maintain a clear dataset with values as accurate as possible.

Duplicate values are handled by just dropping them as they provide no real use to the dataset.

Data normalization is the process of structuring and formatting data to eliminate redundancies, ensure consistency, and bring numeric values into a common scale. In our case, we had to handle the issue of 'height' being in different units. Hence, we separated the value from the unit using the `.split()` function and convert all the data to a single unit(in this case 'cm') so that it's easier to compare values and compute them(if necessary).

Validation refers to the final check required before writing the cleaned data to another source.

The function which plays a crucial role in this is `all()` function which returns true only if all the conditions are fulfilled which is provided in the argument. This provides an effective way as we compile all the checks as a Boolean return into a list and pass it as argument to the `all()` function to check for the validity of the data.

Visualization

In task 5, we were asked to use the cleaned data and visualize them by comparing crucial aspects such as average score by domain where upon generating we can observe that electronics has the highest score validating the fact that students part of the electronics task phase have a higher probability of scoring the most. The next comparison was of attendance vs score so as to check if regularly attending classes translated to a better score. After generating the graph, we observe a linear increasing trend with attendance vs score, stating that there's an effect of good attendance to score well while it's not the only criteria. The last graph was to see the count of submission to check if the majority lies on the yes/no side. We can see that the majority of the students did submit the said project.

Conclusion

Overall, across these 5 tasks, we completely cover the basics of data analysis and data visualization using python by incorporating version control and repositories for future use projects. We make use of the different functions that the python library allows us to use to manipulate and edit data to make them as perfect as possible while also allowing us to add a visual context to the said data which makes it exponentially easier for other users to interpret. We also learnt what version control is, why is it necessary in this context and how to maintain clean and detailed repositories.

Table 1 — Summary of Tasks 1 to 5

Task	Title	Main concept
Task 1	Environment setup and repo structure	Git, virtual environments, .gitignore, requirements.txt
Task 2	CSV to JSON student data analyzer	File I/O, dictionaries, loops, summary generation
Task 3	Manual CSV parser vs pandas	Manual parsing, type conversion, pandas DataFrames
Task 4	Data cleaning	Missing values, duplicates, unit normalization, validation
Task 5	Matplotlib visualization	Bar charts, scatter plots, savefig, axis labels

Figure 1: Tasks and main concepts covered in the Synergy Taskphase

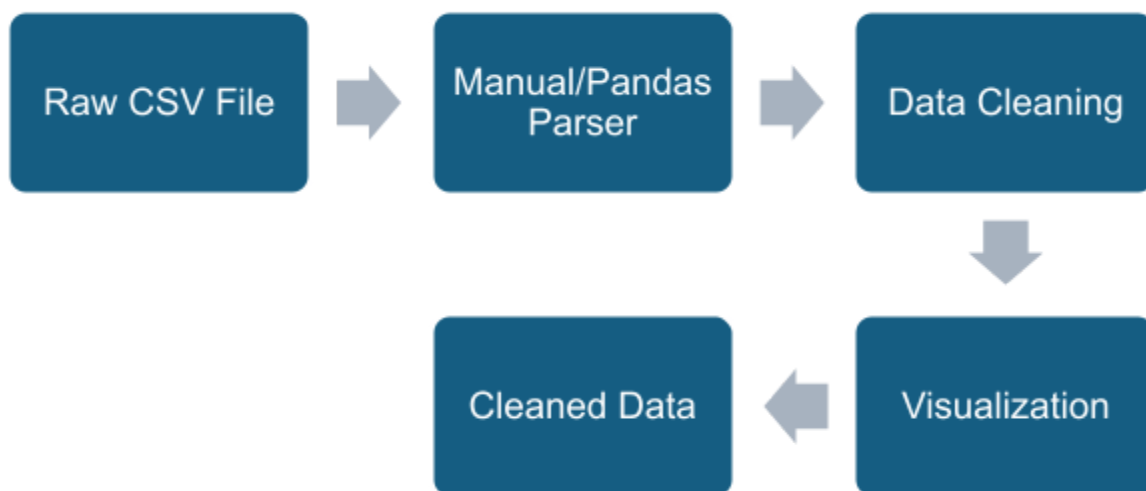


Figure 2: General Workflow of Sequence of events leading to cleaning and visualization of raw data

References:

- 1) Python Documentation: <https://docs.python.org/3/>
- 2) Pandas Documentation: <https://pandas.pydata.org/docs/reference/index.html>
- 3) NumPy Documentation: <https://numpy.org/doc/stable/reference/>
- 4) Matplotlib Documentation: <https://matplotlib.org/stable/api/index.html>
- 5) Direct Google use for certain syntax and example
- 6) Git Tutorial: <https://www.youtube.com/watch?v=Ez8F0nW6S-w>
- 7) Manual Parser(idea): <https://www.youtube.com/watch?v=q5uM4VKywbA>